Multi-Agent Travel Planning System Architecture Flow Document

Document Title: Multi-Agent Travel Planning System with Supervisor Agent, Guardrails & Human-in-the-Loop

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1. Overview

The **Multi-Agent Travel Planning System** is an AI-powered architecture that automates travel planning using multiple specialized AI agents coordinated by a **Supervisor Agent**. The system validates every request through security guardrails, delegates tasks to domain-specific agents, maintains a shared state for collaboration, incorporates human approval, and stores conversations for long-term memory.

This architecture ensures:

- Intelligent task orchestration
- Safe request handling
- Modular agent collaboration
- Human approval before final output
- Persistent memory for future interactions

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2. System Workflow

```
User
 │
 ▼
Input Guardrail
 │
 ├── Invalid Request → Block Request
 │
 ▼
```

```
Supervisor Agent
   |
   ▼
Specialized AI Agents
   |
   ▼
Shared State
   |
   ▼
Human Review
   |
   ├── Request Changes
   |         ▲
   |         |
   └─────────────┘
   |
   ▼
Final Response Agent
   |
   ▼
Persistence & Long-Term Memory
```

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3. Workflow Description

Step 1 – User Input

The workflow begins when the user submits a travel-related request.

Example

> "Plan a 4-day trip to Dubai next month."

The request is forwarded to the Input Guardrail.

Step 2 – Input Guardrail

The Input Guardrail validates every incoming request before processing.

Responsibilities

- Validate user input
- Check request relevance
- Verify security policies
- Detect harmful content
- Prevent invalid workflows

Outcomes

PASS

The request proceeds to the Supervisor Agent.

BLOCK

The system returns an appropriate error message and terminates the workflow.

Step 3 – Supervisor Agent

The Supervisor Agent acts as the central orchestrator.

Responsibilities

- Understand user intent
- Identify required specialist agents
- Create execution strategy
- Manage task delegation
- Coordinate information flow

Output

- Selected agents
- Trip constraints
- Execution reasoning

The Supervisor dynamically determines which agents are required instead of relying on predefined workflows.

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4. Specialist AI Agents

Multiple domain-specific AI agents perform independent tasks.

4.1 Flight Agent

Responsibilities

- Search available flights
- Compare airlines
- Retrieve fares
- Find optimal routes
- Estimate travel duration

Example MCP

AviationStack MCP

4.2 Hotel Agent

Responsibilities

- Search hotels
- Compare prices
- Evaluate amenities
- Analyze customer reviews
- Recommend accommodations

Example MCP

Tavily MCP

4.3 Weather Agent

Responsibilities

- Retrieve weather forecasts
- Analyze climate
- Suggest packing recommendations
- Detect seasonal conditions

Example MCP

Custom Weather MCP

4.4 Budget Agent

Responsibilities

- Estimate travel costs
- Analyze budget constraints
- Recommend spending allocation
- Suggest cost optimization

Example Model

LLM (Llama 3 via MCP)

4.5 Itinerary Agent

Responsibilities

- Generate day-wise itinerary
- Recommend attractions
- Plan transportation
- Schedule activities
- Optimize travel timeline

Example Model

LLM (Llama 3 via MCP)

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5. Shared State (TravelState)

All agents communicate through a centralized shared state.

Stored Information

- User Query
- Trip Constraints
- Flight Results
- Hotel Results
- Weather Information
- Budget Analysis
- Itinerary Plan
- Messages
- LLM Calls

Benefits

- Eliminates duplicate processing
- Enables cross-agent collaboration
- Maintains execution context
- Simplifies orchestration

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6. Human-in-the-Loop

Before delivering the final itinerary, human approval is required.

Options

Approve

The workflow continues to the Final Response Agent.

Request Changes

Requested modifications are sent back to the relevant specialist agents for refinement.

This feedback loop continues until approval is granted.

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7. Final Response Agent

Once approved, the Final Response Agent consolidates all validated information.

Responsibilities

- Merge outputs from all agents
- Generate structured travel plan
- Produce user-friendly recommendations
- Ensure consistency and completeness

Final Output Includes

- Flight recommendations
- Hotel recommendations
- Weather summary
- Budget estimation
- Daily itinerary
- Travel tips

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8. Persistence & Long-Term Memory

The system stores execution history for future reference.

Storage Components

PostgreSQL

- Conversation History
- User Sessions
- State Checkpoints

Shared State

- Cross-agent context
- Workflow progress
- Intermediate results

Benefits

- Conversation continuity

- Memory across sessions
- Faster future planning
- Auditability
- Recovery from failures

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9. Complete Data Flow

```
User
 │
 ▼
Input Guardrail
 │
 ├── Block Invalid Request
 │
 ▼
Supervisor Agent
 │
 ▼
────────────────────────────────────────────
│ Flight Agent                │
│ Hotel Agent                 │
│ Weather Agent               │
│ Budget Agent                │
│ Itinerary Agent             │
────────────────────────────────────────────
 │
 ▼
Shared State (TravelState)
 │
 ▼
Human Review
 │
 ├── Request Changes ────────────────┐
 │                   │
 └───────────────────▶ Specialist Agents ◀┘
 │
 ▼
Final Response Agent
 │
 ▼
PostgreSQL + Long-Term Memory
```

10. Key Features

- Input validation using Guardrails
- Dynamic agent selection through Supervisor Agent
- Modular multi-agent architecture
- Shared state for efficient collaboration
- Human approval before final response
- Persistent memory using PostgreSQL
- Scalable and extensible design
- Improved safety, reliability, and maintainability

Conclusion

The **Multi-Agent Travel Planning System** combines intelligent orchestration, specialized AI agents, centralized state management, human oversight, and persistent memory to deliver accurate, secure, and personalized travel plans. This architecture is highly scalable and can be extended with additional specialist agents such as Visa Agent, Currency Agent, Insurance Agent, Local Guide Agent, or Restaurant Recommendation Agent without affecting the overall workflow.

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